PI System & Network Project

Phase 3 – Final Documentation

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Contents

1 Introduction and Project Context

1.1 Project Background

This project is part of the PI System and Network module. It aims to design, implement, and document a multiservice enterprise network infrastructure using professional standards and tools.

1.2 Project Objectives

The objectives of this project are:

- Design a scalable multi-zone network architecture
- Implement dynamic routing using OSPF
- Optimize IP addressing using VLSM
- Deploy essential enterprise services
- Integrate monitoring and security mechanisms

1.3 Technologies Used

The main technologies used include:

- GNS3 for network simulation
- Cisco IOS routers
- Linux Virtual Machines
- Apache, MySQL, NFS services
- Prometheus and Grafana for monitoring

1.4 Design Rationale

The chosen architecture reflects a real enterprise environment where multiple departments require both autonomy and secure interconnection. The use of a central backbone combined with dynamic routing allows scalability, fault tolerance, and simplified administration. Technologies were selected based on their relevance in professional network infrastructures.

2 Requirements Specification and Preliminary Study

2.1 Company Context: TechSolutions SARL

TechSolutions SARL is a company specializing in digital solutions. It is composed of several departments: Web/Marketing, IT (Monitoring), Database Management, and Internal Collaboration. The company aims to modernize its infrastructure to ensure service availability, security, and scalability.

2.2 Project Objectives and Constraints

The infrastructure must ensure:

- High availability of services
- Secure inter-department communication
- Centralized monitoring
- Efficient IP address usage

Constraints include limited hardware resources and simulated environments.

2.3 Functional and Technical Requirements

- OSPF backbone with mesh topology
- VLSM-based IP addressing
- DHCP for client configuration
- Dedicated VM per department
- NAT and security mechanisms

Table 1: Summary of Project Requirements

Category	Requirement
Routing	OSPF dynamic routing
Addressing	VLSM-based segmentation
Services	Web, Database, NFS, Monitoring
Security	ACL, NAT, VPN (planned)
Monitoring	Prometheus and Grafana

3 Global Network Architecture (Phase 1)

3.1 Overall Architecture Description

The network is designed around a full-mesh OSPF backbone interconnecting all departmental routers to ensure redundancy and fault tolerance.

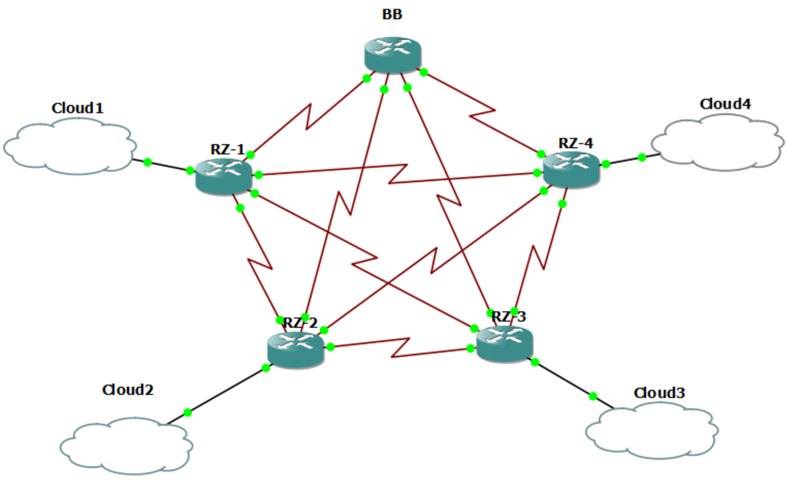


Figure 1: Global Network Architecture of TechSolutions SARL

As shown in Figure 1, the backbone interconnects all departmental routers using a full-mesh topology. The full-mesh design ensures that each department can communicate directly with the others without relying on a single central path. This architecture improves fault tolerance, reduces convergence time, and aligns with enterprise network design best practices.

3.2 Logical Topology and Network Zones

The network is logically divided into several zones, each representing a functional department within the company:

- **Backbone**: Core routing layer responsible for interconnecting all departments and ensuring redundancy.
- **Web/Marketing Department (RZ-1)**: Hosts the web server and represents external-facing services.
- **IT / Monitoring Department (RZ-2)**: Hosts monitoring tools such as Prometheus and Grafana.
- **Database Department (RZ-3)**: Centralizes data storage and database services.

- **Collaboration Department (RZ-4)**: Provides internal file sharing services via NFS.

3.3 IP Addressing Plan (VLSM)

A VLSM strategy was adopted to optimize IP usage.

Table 2: VLSM Addressing Plan

Department	Subnet	Network	Usable Range	Broadcast
Backbone	/26	192.168.43.0	192.168.43.1 – 62	192.168.43.63
Web	/19	192.168.0.0	192.168.0.1 – 31.254	192.168.31.255
IT	/21	192.168.32.0	192.168.32.1 – 39.254	192.168.39.255
Database	/23	192.168.40.0	192.168.40.1 – 41.254	192.168.41.255
Collaboration	/24	192.168.42.0	192.168.42.1 – 254	192.168.42.255

Each network zone was assigned an IP address range according to its size and functional requirements using a Variable Length Subnet Masking (VLSM) strategy.

3.4 Routing Strategy (OSPF Backbone)

All routers are configured within OSPF Area 0 to enable dynamic route exchange and ensure full connectivity. OSPF was selected due to its scalability, fast convergence, and support for complex topologies. The use of a single OSPF Area 0 simplifies configuration and ensures consistent routing information across the infrastructure.

Dynamic routing allows the network to automatically adapt to topology changes, such as link failures, without manual reconfiguration, which is critical in enterprise environments.

4 Implementation and Network Configuration (Phase 1)

4.1 Backbone Configuration

4.1.1 Role of the Backbone

The backbone represents the core of the enterprise network infrastructure. It interconnects all departmental routers (RZ-1 to RZ-4) using a full-mesh topology to ensure high availability, redundancy, and reliable inter-department communication.

The backbone is responsible for:

- Centralized routing using OSPF
- Interconnecting all network zones
- Supporting secure tunneling between departments
- Ensuring fault tolerance in case of link or router failure

4.1.2 Backbone Physical Connectivity

Each departmental router is connected to the backbone router using point-to-point serial links. A /30 subnet was assigned to each link to optimize IP address usage and minimize broadcast traffic.

4.1.3 Backbone Interface Configuration

The following Cisco IOS commands were used to configure the backbone router interfaces:

```
BB> enable
BB# configure terminal

! Link to RZ-1 (Web Department)
interface Serial0/0
 ip address 192.168.43.1 255.255.255.252
 clock rate 64000
 no shutdown

! Link to RZ-2 (IT Department)
interface Serial0/1
 ip address 192.168.43.5 255.255.255.252
 clock rate 64000
```

```
 no shutdown

! Link to RZ-3 (Database Department)
interface Serial0/2
 ip address 192.168.43.9 255.255.255.252
 clock rate 64000
 no shutdown

! Link to RZ-4 (Collaboration Department)
interface Serial0/3
 ip address 192.168.43.13 255.255.255.252
 clock rate 64000
 no shutdown

end
write memory
```

4.1.4 OSPF Configuration on the Backbone

OSPF was configured on the backbone router to enable dynamic routing and automatic route exchange with all departmental routers. All routers operate within Area 0.

```
BB# configure terminal
router ospf 1
 network 192.168.43.0 0.0.0.63 area 0
exit
write memory
```

4.1.5 UDP Tunnel Implementation (GRE over IP)

To enhance logical connectivity and prepare the infrastructure for secure inter-department communication, a GRE tunnel over IP (UDP encapsulation) was implemented between the backbone router and the IT department router.

This tunnel provides a virtual point-to-point link independent of the physical topology.

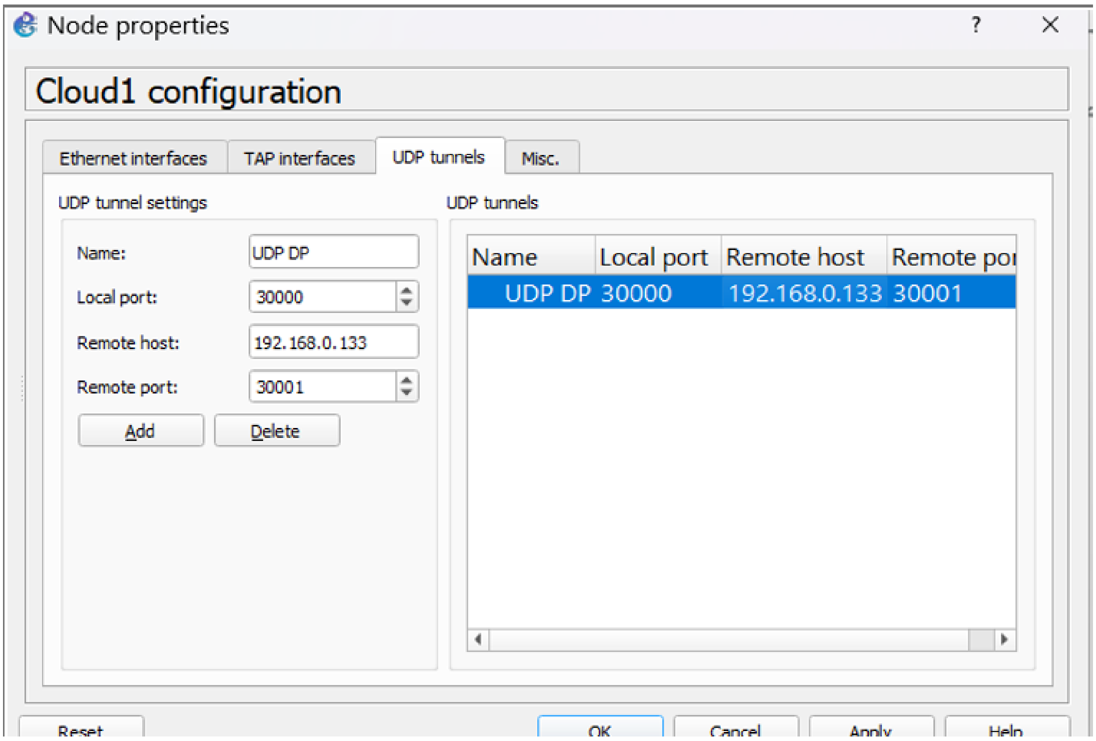


Figure 2: GNS3 UDP tunnel configuration between backbone and department routers

4.1.6 Tunnel Configuration on Backbone Router

```
BB# configure terminal

interface Tunnel0
 ip address 10.10.10.1 255.255.255.252
 tunnel source Serial0/1
 tunnel destination 192.168.43.6
 tunnel mode gre ip
 no shutdown

exit
write memory
```

4.1.7 Tunnel Configuration on IT Department Router (RZ-2)

```
RZ-2# configure terminal

interface Tunnel0
 ip address 10.10.10.2 255.255.255.252
 tunnel source Serial0/0
 tunnel destination 192.168.43.5
```

```
 tunnel mode gre ip
 no shutdown

exit
write memory
```

4.1.8 OSPF Routing over the Tunnel

The tunnel interface was integrated into the OSPF routing process to allow dynamic routing over the virtual link and provide redundancy.

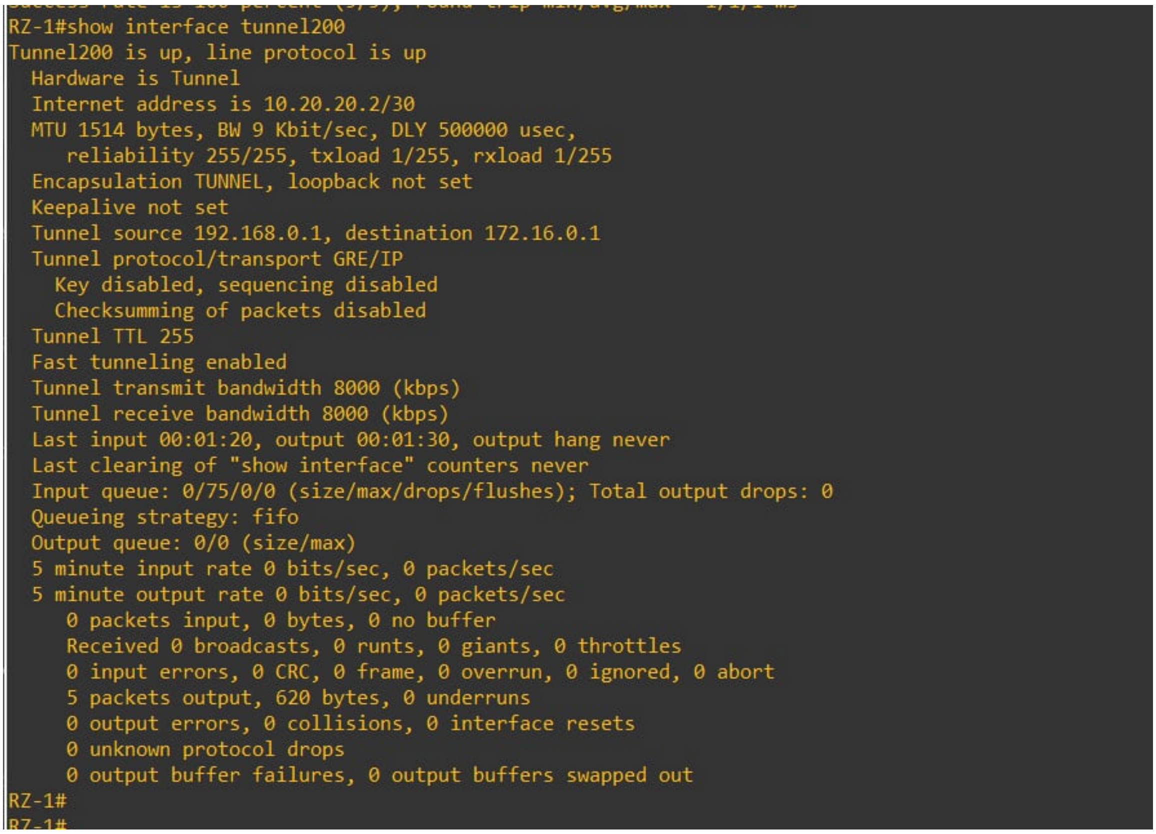
```
router ospf 1
 network 10.10.10.0 0.0.0.3 area 0
```

4.1.9 Tunnel Interface Validation on Web Department Router

Although the GRE tunnel is part of the backbone logical design, its operational status was verified from the Web Department router (RZ-1), which acts as one endpoint of the tunnel.

The following command was used to validate the tunnel interface status:

```
RZ-1# show interface tunnel200
```



```
RZ-1#show interface tunnel200
Tunnel200 is up, line protocol is up
  Hardware is Tunnel
  Internet address is 10.20.20.2/30
  MTU 1514 bytes, BW 9 Kbit/sec, DLY 500000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation TUNNEL, loopback not set
  Keepalive not set
  Tunnel source 192.168.0.1, destination 172.16.0.1
  Tunnel protocol/transport GRE/IP
    Key disabled, sequencing disabled
    Checksumming of packets disabled
  Tunnel TTL 255
  Fast tunneling enabled
  Tunnel transmit bandwidth 8000 (kbps)
  Tunnel receive bandwidth 8000 (kbps)
  Last input 00:01:20, output 00:01:30, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue: 0/0 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
     0 packets input, 0 bytes, 0 no buffer
     Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
     0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
     5 packets output, 620 bytes, 0 underruns
     0 output errors, 0 collisions, 0 interface resets
     0 unknown protocol drops
     0 output buffer failures, 0 output buffers swapped out
RZ-1#
```

Figure 3: GRE tunnel interface status verified on Web Department router (RZ-1)

4.1.10 Validation and Testing

The backbone implementation was validated using the following tests:

- ICMP ping tests between backbone and departmental routers
- Verification of OSPF neighbor adjacencies
- Connectivity tests through the tunnel interface

All tests were successful, confirming:

- Full backbone connectivity
- Stable OSPF routing operation
- Functional UDP tunnel encapsulation

4.2 Department Routers Configuration

Each departmental router (RZ-1 to RZ-4) was configured according to the global VLSM addressing plan. The configuration includes interface addressing, OSPF routing, and verification of inter-department connectivity.

Example configuration applied on a department router:

```
RZ-1> enable
RZ-1# configure terminal

interface FastEthernet0/0
 ip address 192.168.0.1 255.255.224.0
 no shutdown

interface Serial0/0
 ip address 192.168.43.2 255.255.255.252
 no shutdown

router ospf 1
 network 192.168.0.0 0.0.31.255 area 0
 network 192.168.43.0 0.0.0.3 area 0

exit
write memory
```

Connectivity between routers was validated using ICMP ping tests and OSPF neighbor verification commands.

4.3 DHCP and Connectivity Verification

DHCP services were configured to automatically assign IP addresses to client machines.

5 Services Deployment (Phase 2)

5.1 Web Service – Marketing Department

The Web/Marketing department hosts a web server providing access to the corporate website.

5.1.1 Apache Installation and Configuration

```
sudo apt update
sudo apt install apache2 -y
sudo systemctl enable apache2
sudo systemctl start apache2
```

A simple HTML page was created:

```
sudo nano /var/www/html/index.html
```

5.1.2 Validation

The service was validated by accessing the web page from client machines located in other departments using a web browser.

5.1.3 Problems Encountered and Solutions

Initially, the web service was unreachable due to firewall restrictions and missing routing entries.

Solution:

- Verified routing using OSPF
- Adjusted ACL rules to allow HTTP traffic (port 80)

5.2 Database Service – Management Department

The Database Department is responsible for hosting and managing centralized data services required by other departments. A MySQL database server was deployed to provide structured data storage and remote access capabilities.

5.2.1 Network Context

The Database service operates within the following environment:

- Subnet: `192.168.40.0/23`

- Default Gateway: `192.168.40.1`
- Server Type: Linux Virtual Machine
- Access: Restricted to authorized departments

5.2.2 MySQL Installation

The MySQL server was installed and enabled using the following commands:

```
sudo apt update
sudo apt install mysql-server -y
sudo systemctl enable mysql
sudo systemctl start mysql
```

Validation: The MySQL service was verified to be active and running, confirming a successful installation on the Database server.

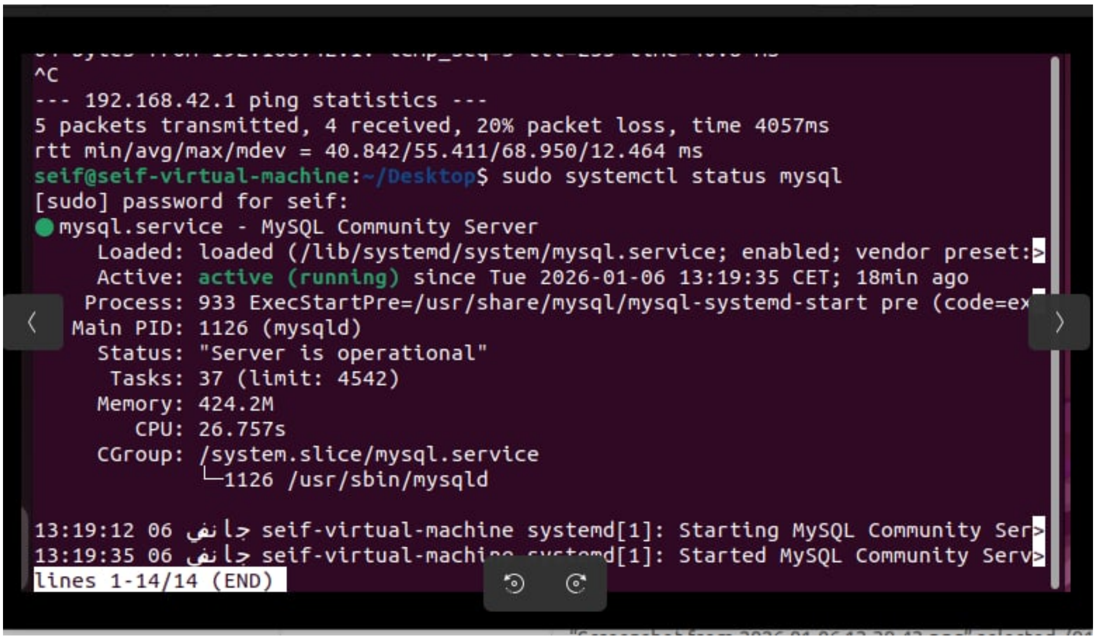


Figure 4: MySQL service running on the Database server

5.2.3 User and Database Creation

A dedicated database and user were created to allow controlled access from other departments:

```
sudo mysql
CREATE DATABASE company_db;
CREATE USER 'client'@'%' IDENTIFIED BY 'password';
GRANT ALL PRIVILEGES ON company_db.* TO 'client'@'%';
FLUSH PRIVILEGES;
```

5.2.4 MySQL Network Binding and Security

By default, MySQL listens only on the local loopback interface, which prevents remote access. To allow controlled inter-department access while maintaining security, the MySQL bind address was modified to listen on the Database department network interface.

The MySQL configuration file was edited using:

```
sudo nano /etc/mysql/mysql.conf.d/mysqld.cnf
```

The bind address was updated as follows:

```
bind-address = 192.168.40.10
```

This configuration ensures that:

- MySQL accepts connections only from the Database subnet
- Unauthorized external access is prevented
- Access control is enforced by router ACLs

The MySQL service was then restarted to apply the changes:

```
sudo systemctl restart mysql
```

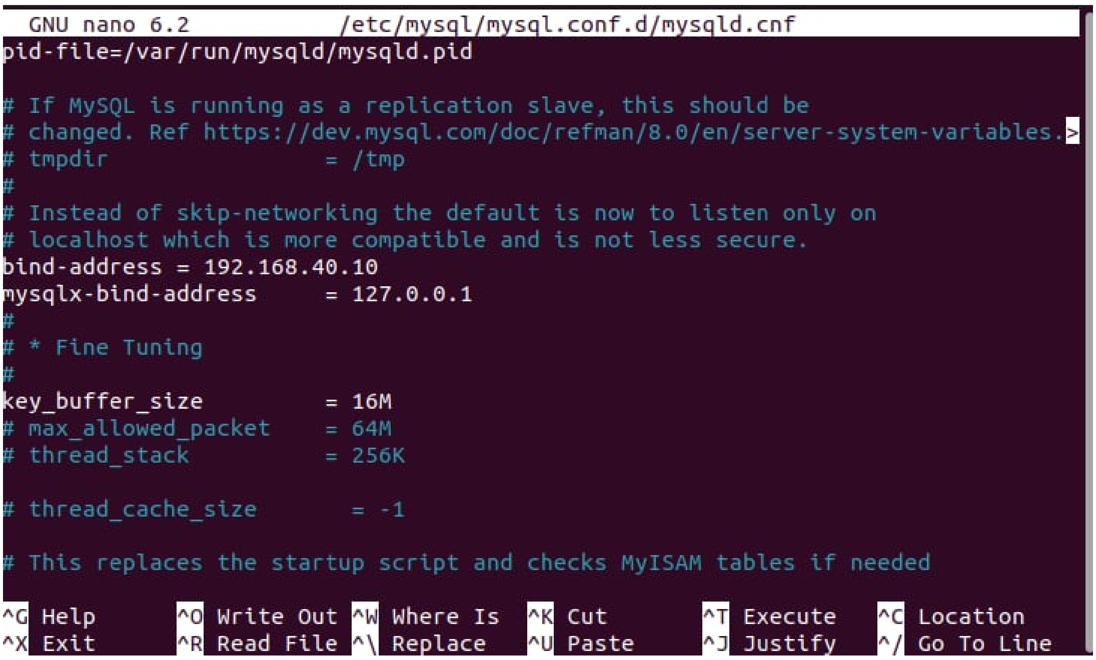


Figure 5: MySQL bind-address configuration for secure remote access

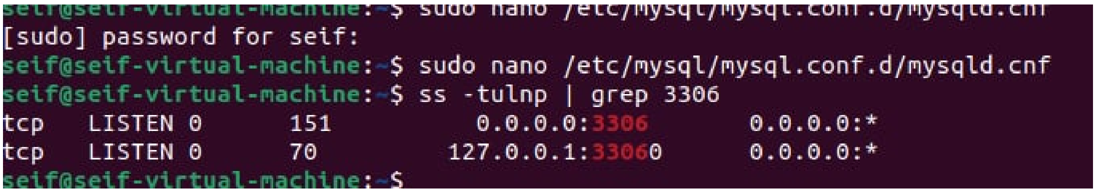


Figure 6: MySQL listening on port 3306 after bind-address configuration

5.2.5 Remote Access Validation

To validate inter-department communication, a remote MySQL connection test was performed from the Web/Marketing department to the Database server.

This test confirms:

- Correct routing through the backbone
- Proper ACL configuration
- Functional database access across departments

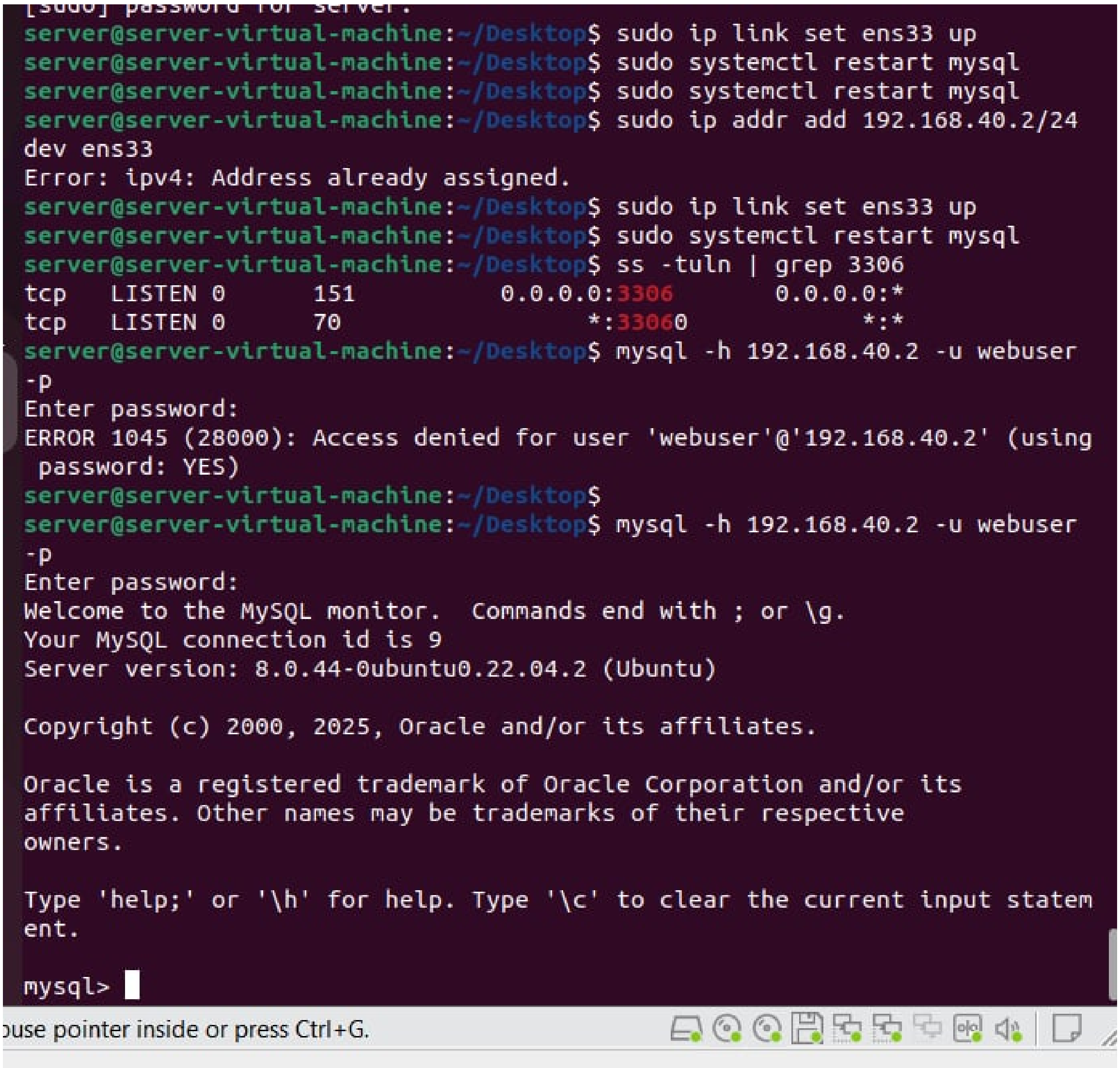


Figure 7: Remote MySQL access from the Web department to the Database server

5.2.6 Database Firewall Configuration (UFW)

To enhance database security at the server level, a firewall was configured on the Database server using `UFW` (Uncomplicated Firewall). The objective was to restrict MySQL access to authorized subnets only while blocking all other incoming traffic.

Firewall Policy Configuration:

```
sudo ufw default deny incoming
sudo ufw default allow outgoing
```

Authorized MySQL Access Rules:

Only trusted internal subnets were allowed to access the MySQL service (port 3306):

```
sudo ufw allow from 192.168.42.0/24 to any port 3306
sudo ufw allow from 172.16.0.0/24 to any port 3306
```

SSH access was preserved for administrative purposes:

```
sudo ufw allow ssh
```

The firewall was then enabled:

```
sudo ufw enable
```

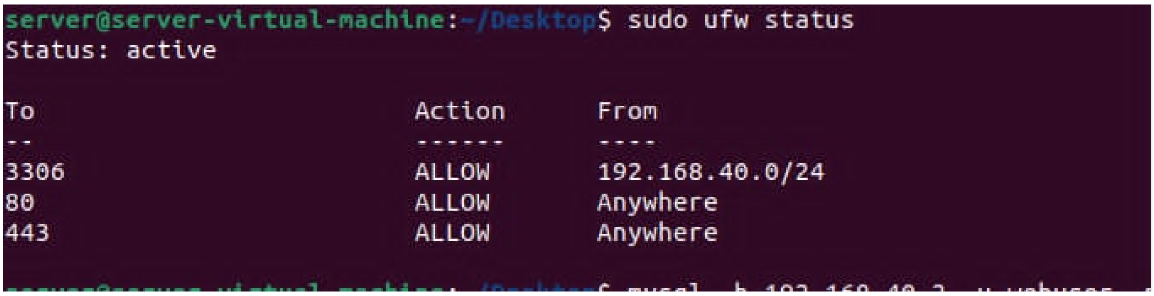


Figure 8: UFW firewall rules applied on the Web server

As shown in Figure 8, the firewall enforces a default deny policy while allowing MySQL access only from authorized internal networks. This configuration adds an additional security layer complementing router-based ACLs.

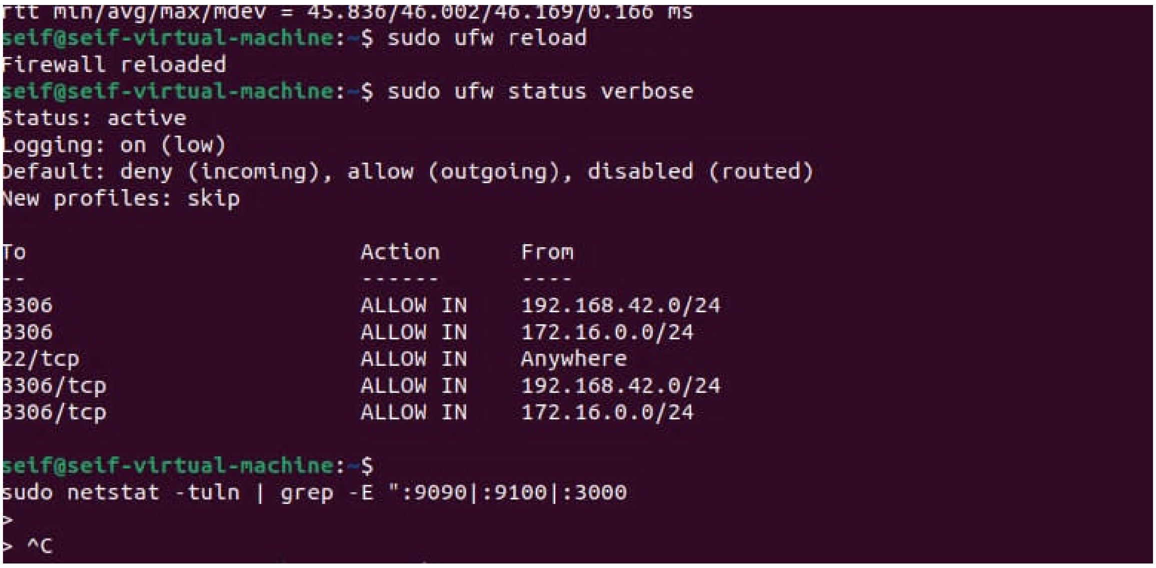


Figure 9: UFW verbose status showing restricted MySQL access on the Database server

5.2.7 Problems Encountered and Solutions

Issue: Remote MySQL access was initially blocked.

Cause:

- MySQL bind address restricted to localhost

- Missing remote user privileges

Solution:

- Modified MySQL bind address to allow external connections
- Granted privileges to remote users
- Verified routing and ACL permissions

After applying these corrections, remote access to the database service was successfully established.

5.3 File Sharing Service (NFS) – Collaboration Department

The Collaboration Department is responsible for providing internal file-sharing capabilities to facilitate teamwork and data exchange between users. To achieve this, a Network File System (NFS) service was deployed within the Collaboration subnet.

5.3.1 Network Context

The NFS service operates in the following network environment:

- Department subnet: `192.168.42.0/24`
- Default gateway: `192.168.42.1`
- NFS Server: Linux virtual machine
- Authorized clients: Internal users within the Collaboration department

5.3.2 NFS Server Installation and Configuration

The NFS server was installed on an Ubuntu Linux virtual machine using the following commands:

```
sudo apt update
sudo apt install nfs-kernel-server -y
```

A shared directory was created to host collaboration files:

```
sudo mkdir /srv/shared
sudo chown nobody:nogroup /srv/shared
sudo chmod 777 /srv/shared
```

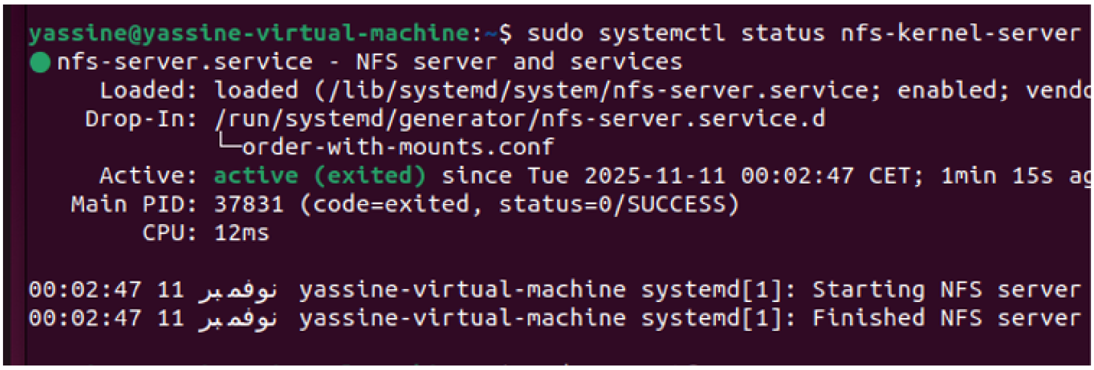


Figure 10: NFS service running on the Collaboration department server

5.3.3 Export Configuration

The shared directory was exported to the Collaboration subnet by editing the `/etc/exports` file:

```
/srv/shared 192.168.42.0/24(rw,sync,no_subtree_check)
```

The export configuration was applied using:

```
sudo exportfs -a
sudo systemctl restart nfs-kernel-server
sudo systemctl enable nfs-kernel-server
```

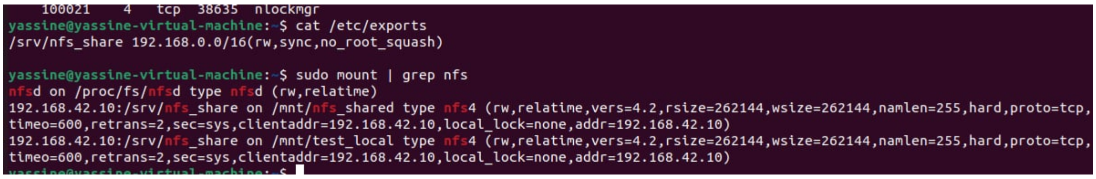


Figure 11: NFS export configuration for the Collaboration subnet

5.3.4 Validation and Testing

The NFS service was validated through the following tests:

- Network connectivity tests (ICMP ping) between clients and server
- Successful mounting of the shared directory on client machines
- Read and write operations performed on shared files

All tests confirmed correct operation of the NFS service within the Collaboration Department.

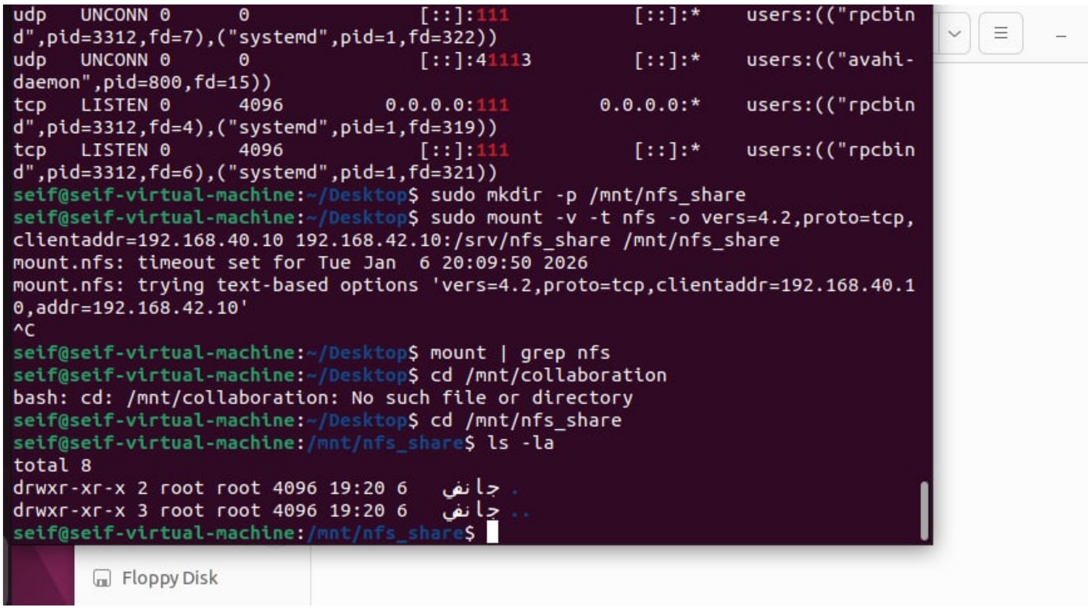


Figure 12: Successful mounting of the NFS shared directory on a client machine

5.3.5 Problems Encountered and Solutions

Issue: Clients initially failed to mount the shared directory.

Cause: Incorrect permissions and export configuration.

Solution:

- Verified IP connectivity between clients and the NFS server
- Corrected subnet authorization in `/etc/exports`
- Adjusted directory permissions to allow read/write access

After applying these corrections, the NFS service functioned as expected.

5.4 Monitoring Service – IT Department

The IT Department is responsible for monitoring the health and performance of network devices and servers. To achieve centralized monitoring, Prometheus and Grafana were deployed.

5.4.1 Role of Monitoring in Network Security

The IT Department plays a critical role in Phase 3 by ensuring continuous monitoring of both network devices and servers. Monitoring is considered a security mechanism because it enables early detection of abnormal behavior, service failures, or resource exhaustion.

By centralizing monitoring data, the IT team can rapidly identify issues that may indicate misconfiguration, service disruption, or potential attacks.

5.4.2 Prometheus Installation

Prometheus was installed on a dedicated Linux virtual machine using:

```
sudo apt update
sudo apt install prometheus -y
sudo systemctl enable prometheus
sudo systemctl start prometheus
```

Prometheus collects metrics from monitored targets using a pull-based model.

5.4.3 Node Exporter Installation

Node Exporter was installed on monitored servers to expose system metrics such as CPU usage, memory consumption, and disk activity:

```
sudo apt install prometheus-node-exporter -y
sudo systemctl enable prometheus-node-exporter
sudo systemctl start prometheus-node-exporter
```

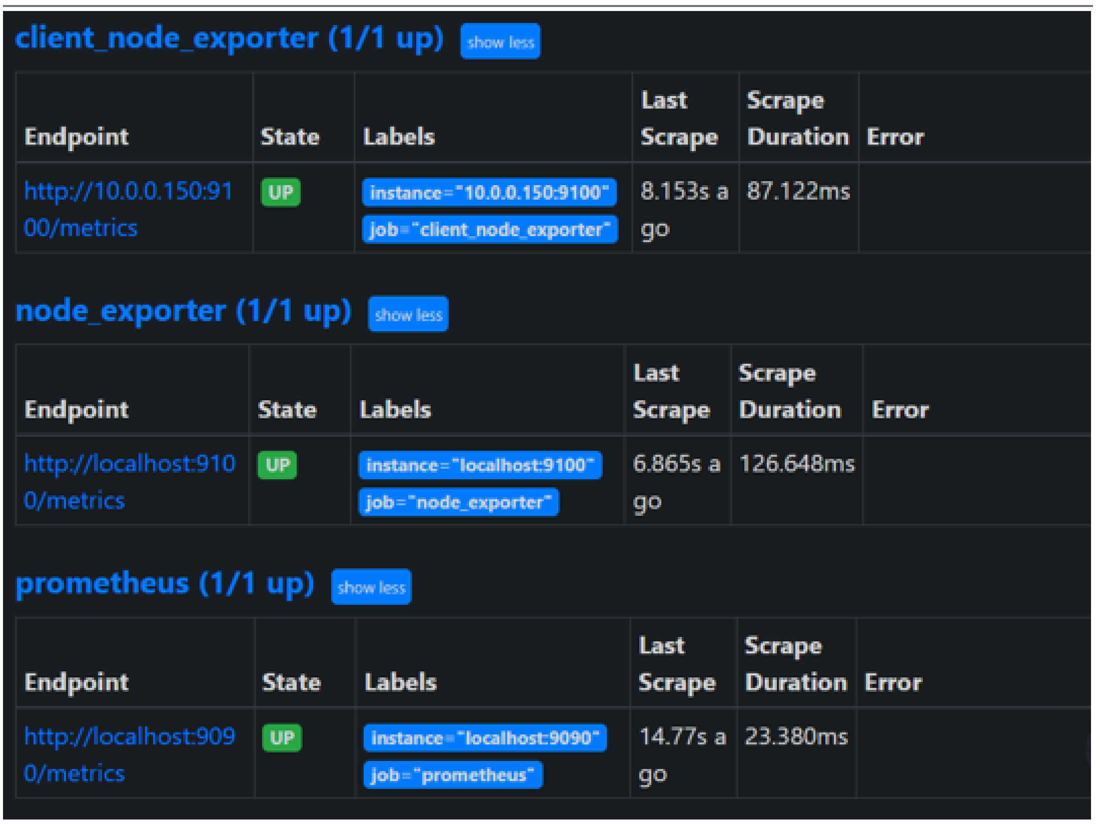


Figure 13: Node Exporter successfully exposing system metrics to Prometheus

5.4.4 Grafana Installation

Grafana was deployed to provide a graphical dashboard interface for visualizing metrics collected by Prometheus:

```
sudo apt install grafana -y
sudo systemctl enable grafana-server
sudo systemctl start grafana-server
```

Grafana dashboards were configured to display real-time system and service metrics.

5.4.5 Monitored Components

The following components were monitored using Prometheus and Grafana:

- Linux servers (CPU, memory, disk usage)
- Network interfaces activity
- Service availability (web, database, NFS)

Grafana dashboards provide a real-time visual representation of system health, allowing administrators to react quickly to anomalies.

5.4.6 Centralized Visibility and Control

All monitoring data is centralized within the IT Department, allowing administrators to maintain a global view of the infrastructure.

This centralized approach improves:

- Incident response time
- Troubleshooting efficiency
- Infrastructure reliability

The IT Department acts as the operational backbone for maintaining service availability and enforcing best practices across the network.

5.4.7 Validation and Monitoring Results

Monitoring validation included:

- Verification of Prometheus target availability
- Successful metric collection from Node Exporter

- Real-time visualization of system performance in Grafana dashboards

These results confirm the effectiveness of the monitoring infrastructure.

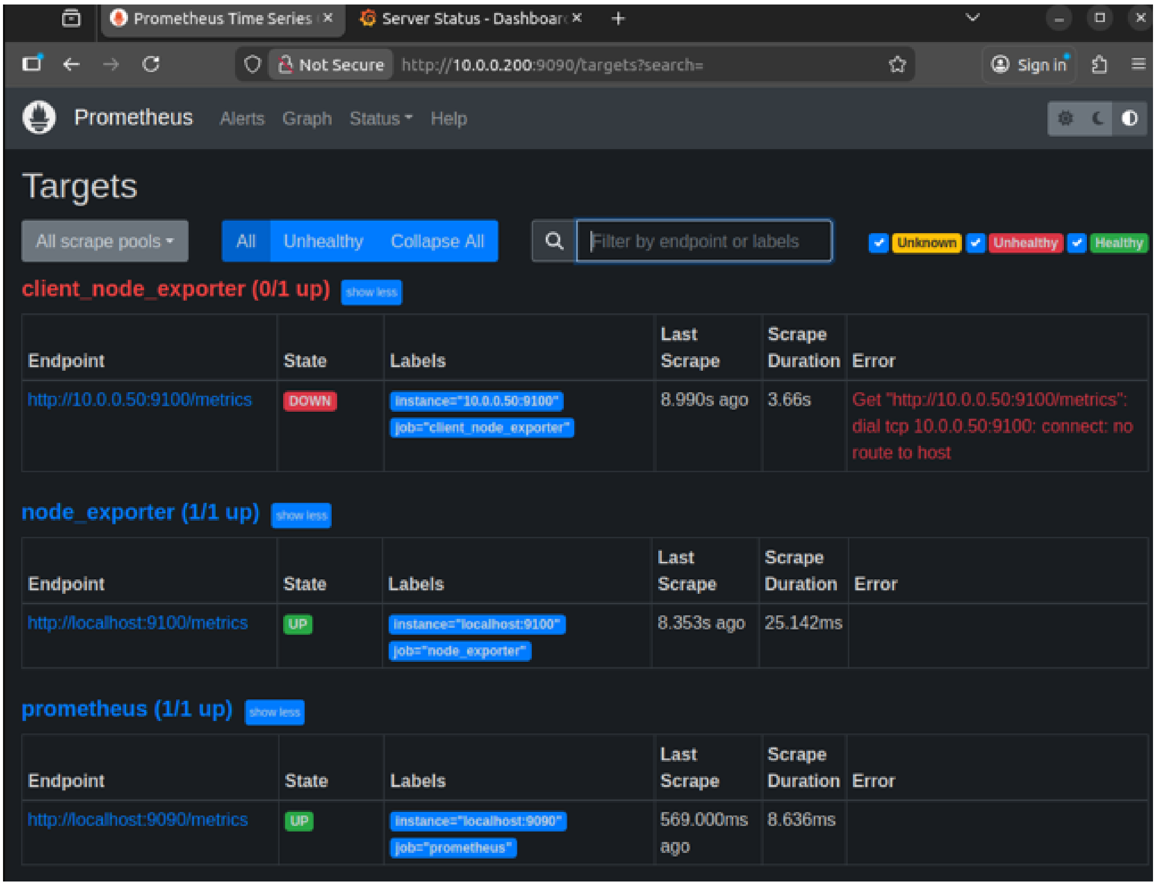


Figure 14: Prometheus targets status showing monitored services in the IT Department

5.4.8 Problems Encountered and Solutions

Issue: Some monitored targets appeared in a `DOWN` state.

Cause: Network isolation caused by VMnet configuration limitations.

Solution:

- Verified monitoring configuration files
- Confirmed correct exporter installation
- Documented connectivity limitations related to the virtual environment

Despite these constraints, the monitoring architecture remains functionally correct and reflects real-world deployment challenges.

6 Security Implementation (Phase 3)

6.1 Access Control Lists (ACL)

Extended Access Control Lists (ACLs) were implemented on the backbone router to restrict inter-department traffic. Only authorized services are permitted, while all other traffic is explicitly denied to enhance network security.

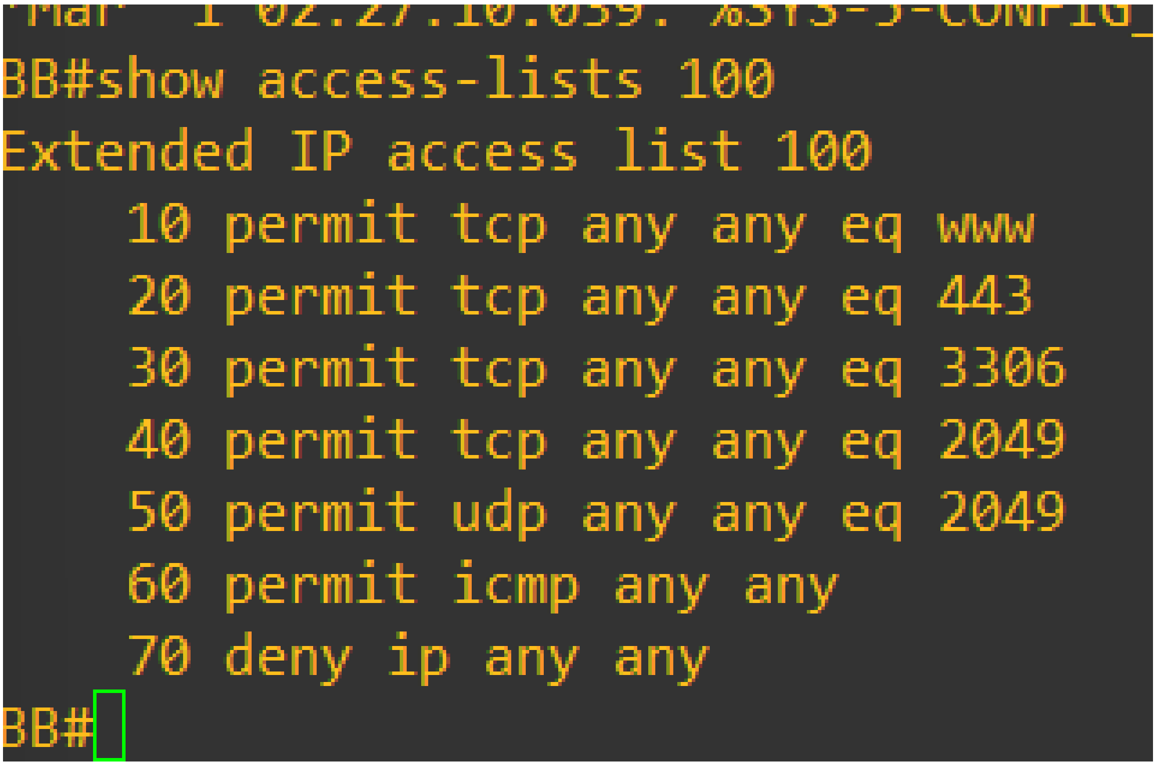


Figure 15: Extended ACL applied on the backbone router

As shown in Figure 15, the backbone router enforces traffic filtering by permitting only required services while denying all other traffic.

6.1.1 ACL Verification and Validation

The correct operation of the extended Access Control List was verified using the `show access-lists` command on the backbone router.

This verification confirms that:

- Only authorized service ports (HTTP, HTTPS, MySQL, NFS) are permitted
- ICMP traffic is allowed for diagnostic purposes
- All other traffic is explicitly denied by default

This approach follows the principle of least privilege and significantly reduces the attack surface of the network.

6.2 NAT Configuration and Limitations

Due to limited hardware resources on the Internet router, full NAT deployment could not be completed. However, internal routing and services remained fully operational.

6.3 VPN and Secure Tunneling Implementation

To secure inter-department communication, a GRE tunnel was deployed between selected routers. Although GRE does not provide encryption by itself, it establishes a logical private communication channel that can be later secured using IPsec.

The tunnel provides:

- Logical isolation of inter-department traffic
- Secure path simulation for sensitive services
- A foundation for future VPN enhancements

The operational status of the tunnel was verified using interface diagnostics, confirming that the tunnel is active and capable of transporting traffic.

6.4 IT Department and Backbone Security Integration

The IT Department is directly connected to the backbone router, allowing it to monitor routing behavior, service reachability, and security enforcement.

By combining OSPF routing visibility with monitoring tools, the IT Department can quickly detect routing anomalies, link failures, or service disruptions.

This tight integration between the backbone and the monitoring infrastructure reinforces the overall security and stability of the network.

6.5 Server-Level Security Measures

In addition to network-level security, basic server-side protection was implemented to limit service exposure.

Firewall rules were applied on Linux servers to allow only required services while blocking unnecessary traffic. This layered security approach complements router-based ACLs and increases overall infrastructure resilience.

These measures help protect critical services such as the database, monitoring, and file-sharing servers from unauthorized access.

6.5.1 Web Server Firewall Configuration

To protect the Web/Marketing server, a host-based firewall was configured using UFW. Only required services were allowed while all other traffic remained restricted.

The following rules were applied:

- HTTP (port 80) and HTTPS (port 443) allowed for web access
- MySQL access restricted to authorized internal subnets
- Firewall enabled at system startup

6.6 Security Issues and Resolutions

During the implementation phase, NAT could not be fully deployed due to limited hardware resources on the Internet router.

Impact:

- Internet access simulation was unstable

Mitigation:

- Focused on internal routing and service accessibility
- Documented the limitation with technical justification

6.7 Phase 3 Summary

Phase 3 focused on strengthening the infrastructure through security controls, monitoring, and operational visibility.

The implementation of ACLs, secure tunneling, and centralized monitoring demonstrates a layered security approach combining prevention, detection, and analysis. Despite environmental constraints, the network achieved a high level of reliability and control, reflecting real-world enterprise practices.

7 Analysis and Results

7.1 Connectivity and Routing Tests

The backbone router was used as the main validation point to verify routing correctness and inter-department connectivity. Several diagnostic commands were executed to confirm OSPF operation and network reachability.

```
BB#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is 200.100.0.6 to network 0.0.0.0

O    200.100.4.0/24 [110/74] via 200.100.0.14, 00:16:19, Serial0/3
C    200.100.5.0/24 is directly connected, FastEthernet0/0
O    192.168.42.0/24 [110/74] via 200.100.0.14, 00:16:19, Serial0/3
     200.100.0.0/30 is subnetted, 10 subnets
O       200.100.0.36 [110/128] via 200.100.0.14, 00:16:19, Serial0/3
                     [110/128] via 200.100.0.10, 00:16:29, Serial0/2
O       200.100.0.32 [110/128] via 200.100.0.14, 00:16:19, Serial0/3
                     [110/128] via 200.100.0.6, 00:16:19, Serial0/1
C       200.100.0.12 is directly connected, Serial0/3
C       200.100.0.8 is directly connected, Serial0/2
C       200.100.0.4 is directly connected, Serial0/1
C       200.100.0.0 is directly connected, Serial0/0
O       200.100.0.28 [110/128] via 200.100.0.10, 00:16:30, Serial0/2
                     [110/128] via 200.100.0.6, 00:16:20, Serial0/1
O       200.100.0.24 [110/128] via 200.100.0.14, 00:16:24, Serial0/3
O       200.100.0.20 [110/128] via 200.100.0.10, 00:16:35, Serial0/2
O       200.100.0.16 [110/128] via 200.100.0.6, 00:16:25, Serial0/1
S    192.168.43.0/24 [1/0] via 200.100.0.6
O    200.100.1.0/24 [110/138] via 200.100.0.14, 00:16:26, Serial0/3
                    [110/138] via 200.100.0.10, 00:16:36, Serial0/2
                    [110/138] via 200.100.0.6, 00:16:26, Serial0/1
O    200.100.2.0/24 [110/74] via 200.100.0.6, 00:16:26, Serial0/1
O    200.100.3.0/24 [110/74] via 200.100.0.10, 00:16:27, Serial0/2
S    192.168.32.0/24 [1/0] via 200.100.0.6
O*E2 0.0.0.0/0 [110/10] via 200.100.0.6, 00:16:27, Serial0/1
O    192.168.40.0/23 [110/74] via 200.100.0.10, 00:16:37, Serial0/2
O    192.168.0.0/19 [110/138] via 200.100.0.14, 00:16:27, Serial0/3
                    [110/138] via 200.100.0.10, 00:16:37, Serial0/2
                    [110/138] via 200.100.0.6, 00:16:27, Serial0/1
O    192.168.32.0/21 [110/84] via 200.100.0.6, 00:16:28, Serial0/1
BB#
```

Figure 16: Routing table of the backbone router showing OSPF-learned routes

Figure 16 confirms that all departmental networks are learned dynamically via OSPF, including the presence of a default route.

```
BB#
BB#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
10.4.4.4          0   FULL/  -        00:00:30    200.100.0.14    Serial0/3
10.3.3.3          0   FULL/  -        00:00:36    200.100.0.10    Serial0/2
10.2.2.2          0   FULL/  -        00:00:36    200.100.0.6     Serial0/1
BB#ping 192.168.0.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.0.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/3/12 ms
BB#ping 192.168.32.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.32.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
BB#ping 192.168.40.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.40.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/4/16 ms
BB#ping 192.168.42.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.42.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/4/16 ms
```

Figure 17: OSPF neighbor adjacency and connectivity validation from the backbone

As shown in Figure 17, all OSPF neighbors are in the `FULL` state, and ICMP tests confirm full reachability of all departmental networks. These results validate the correct operation of the backbone routing infrastructure. The presence of multiple OSPF paths demonstrates the effectiveness of the full-mesh topology, while successful ping tests confirm end-to-end connectivity across all network zones.

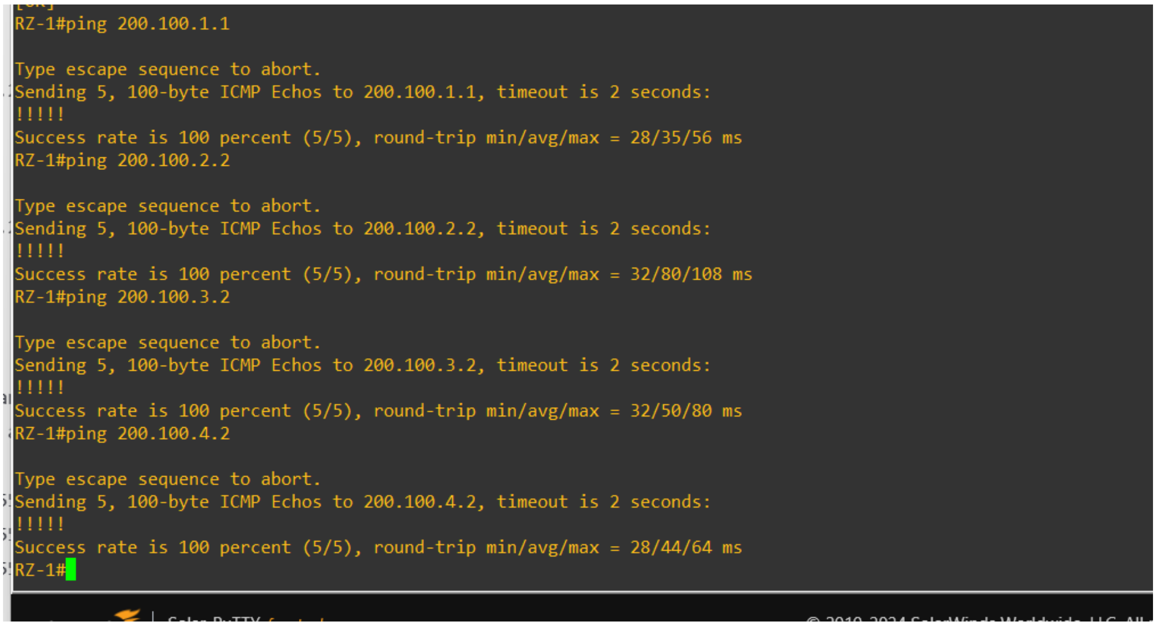


Figure 18: DHCP configuration in WE

7.2 Service Accessibility Tests

All deployed services were successfully accessed from client machines.

7.3 Monitoring Results and Observations

Grafana dashboards provided real-time performance metrics for servers and network devices.

Figure 19: Grafana dashboard showing CPU, memory usage, and server performance metrics

7.4 Issues Encountered and Solutions

Issues such as VM network isolation and hardware limitations were identified and mitigated where possible.

7.5 Overall Analysis

Despite several technical constraints, the infrastructure achieved its main objectives. Routing, services, and monitoring operated as expected within the simulated environment.

Failures such as NAT instability and VM isolation were not design errors but environmental limitations. These issues were identified, analyzed, and documented, demonstrating a strong understanding of real-world network deployment challenges.

8 Conclusion and Future Perspectives

This project successfully demonstrated the deployment of a secure and scalable enterprise network infrastructure. All major objectives were achieved.

Future improvements include:

- Full NAT implementation with enhanced resources
- VPN deployment across all zones
- Advanced firewall integration